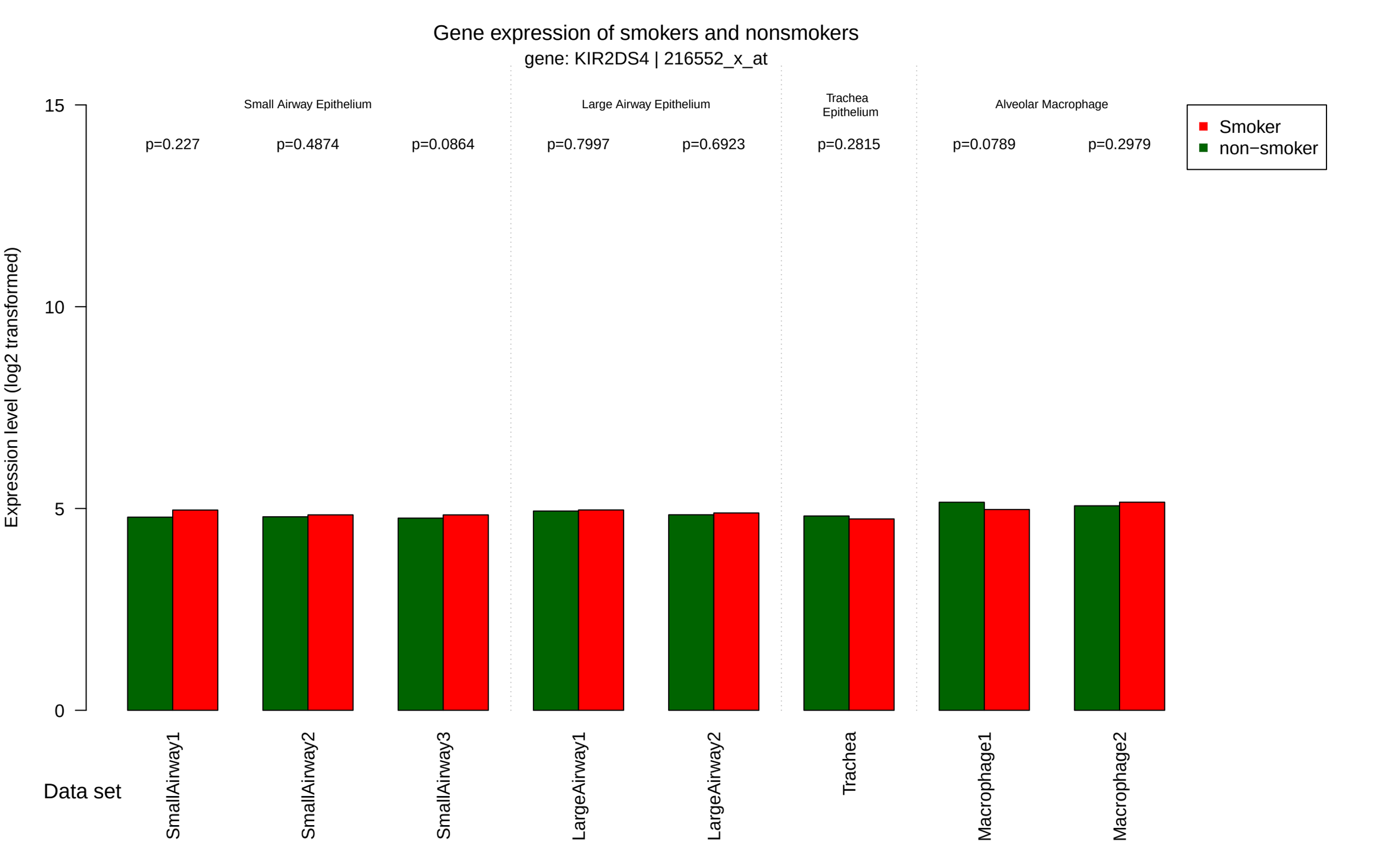
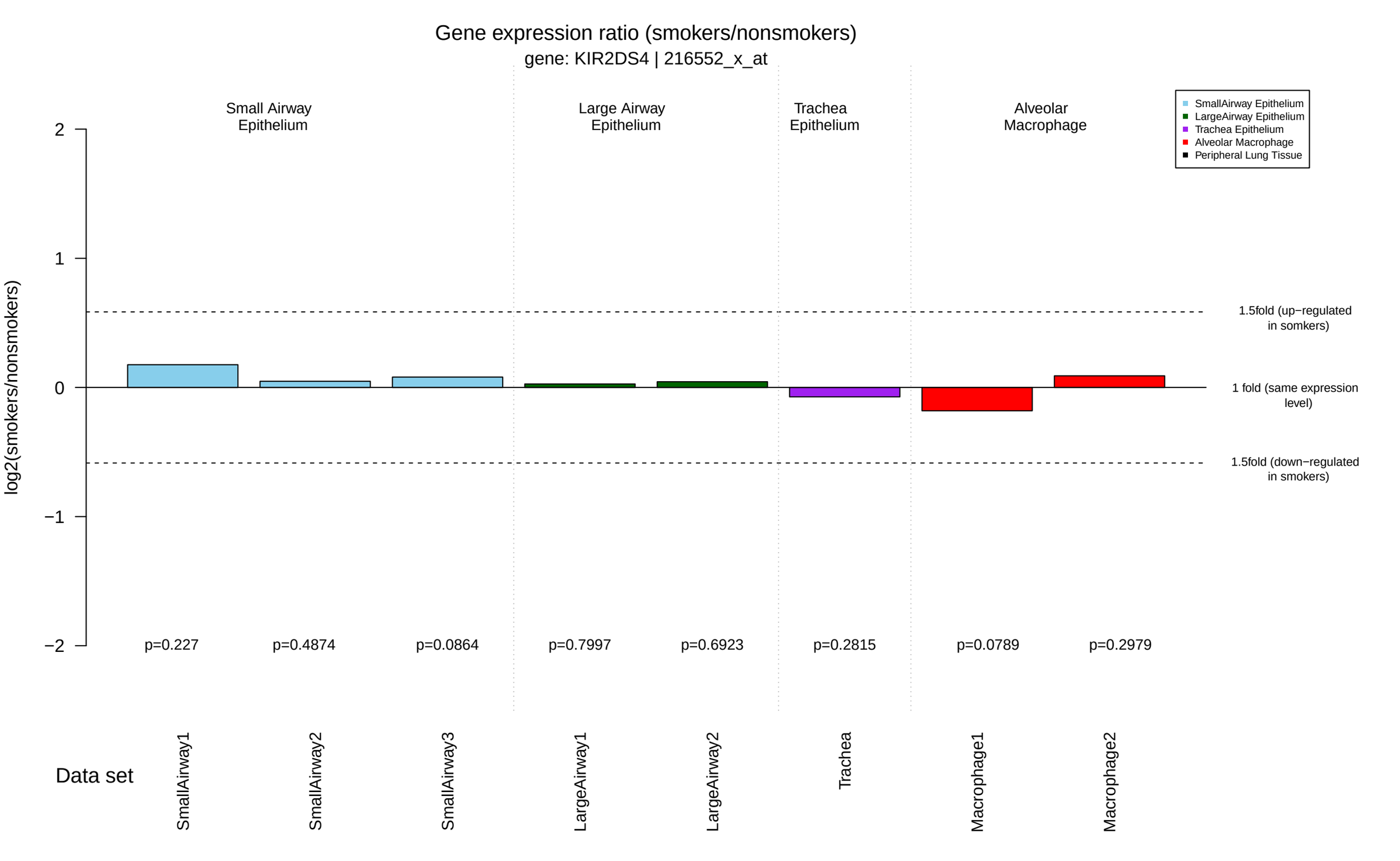
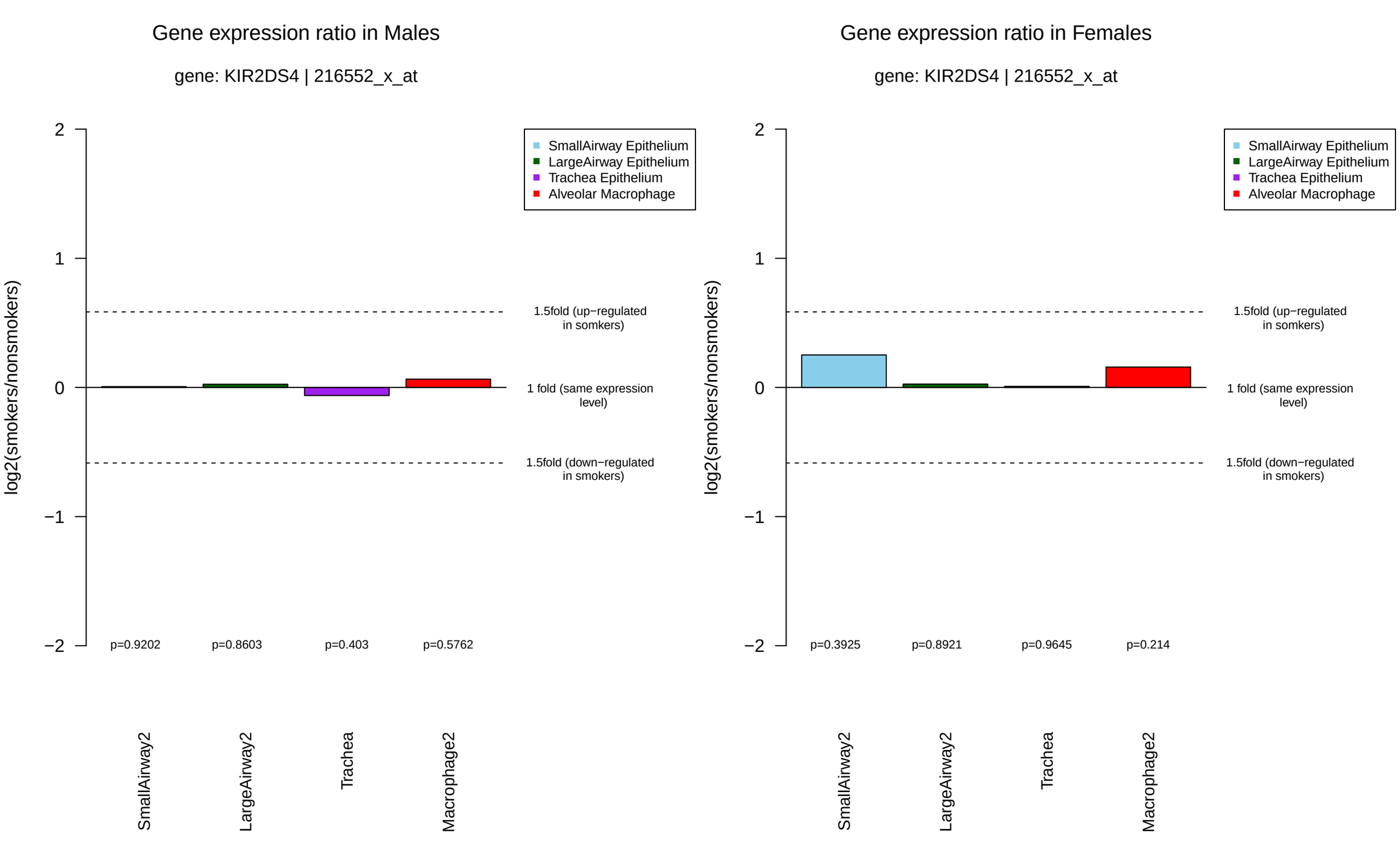
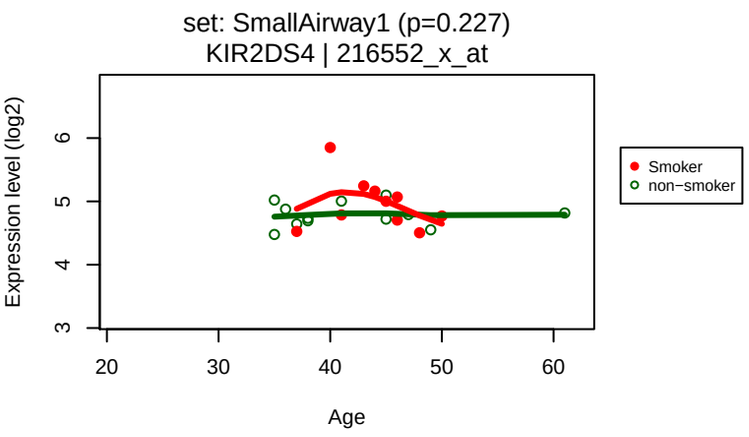
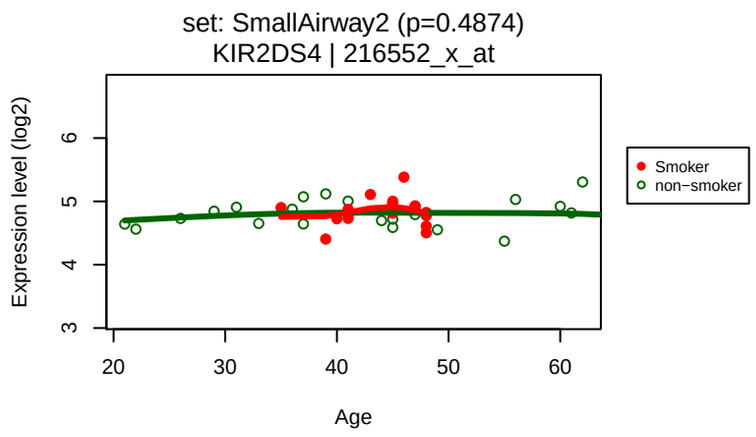
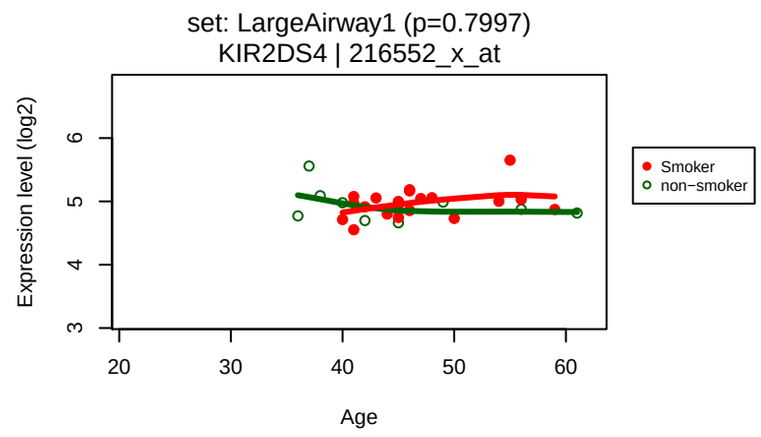

Gene expression of smokers and nonsmokers
gene: KIR2DS4 | 216552_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.227
p=0.4874
p=0.0864
p=0.7997
p=0.6923
p=0.2815
p=0.0789
p=0.2979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

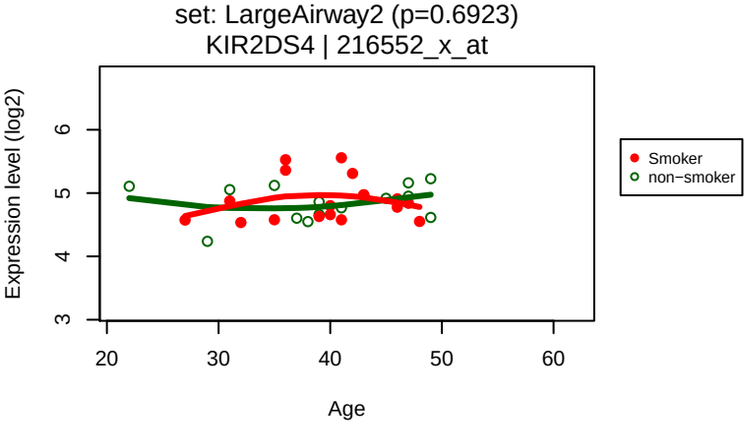

Gene expression ratio (smokers/nonsmokers)
gene: KIR2DS4 | 216552_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.227
p=0.4874
p=0.0864
p=0.7997
p=0.6923
p=0.2815
p=0.0789
p=0.2979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

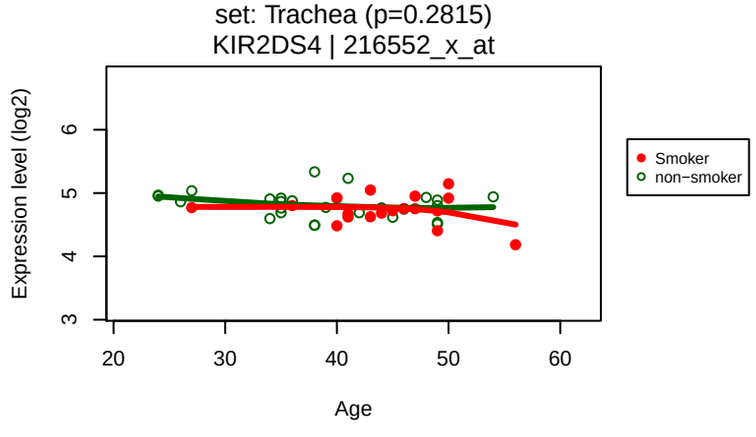

Gene expression ratio in Males
gene: KIR2DS4 | 216552_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9202
p=0.8603
p=0.403
p=0.5762
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KIR2DS4 | 216552_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3925
p=0.8921
p=0.9645
p=0.214
SmallAirway2
LargeAirway2
Trachea
Macrophage2

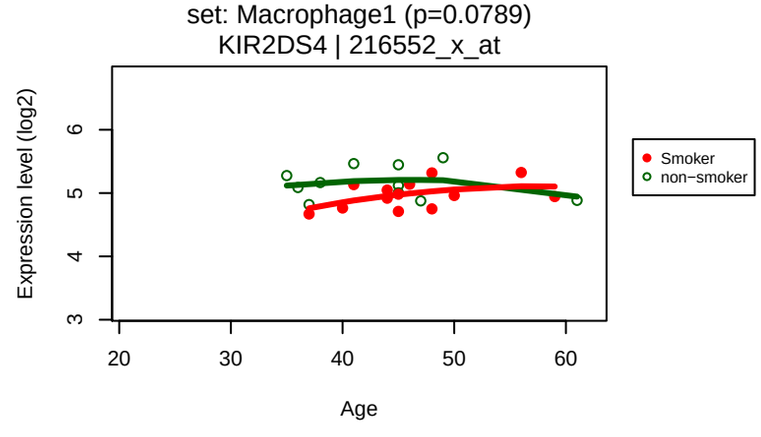

set: SmallAirway1 (p=0.227)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker

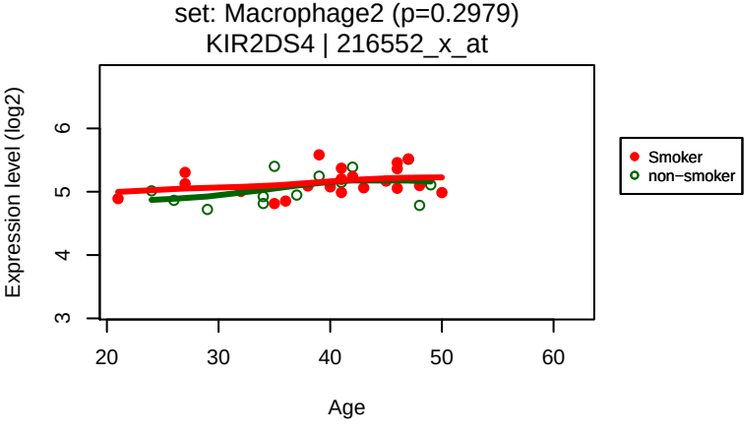

set: SmallAirway2 (p=0.4874)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker

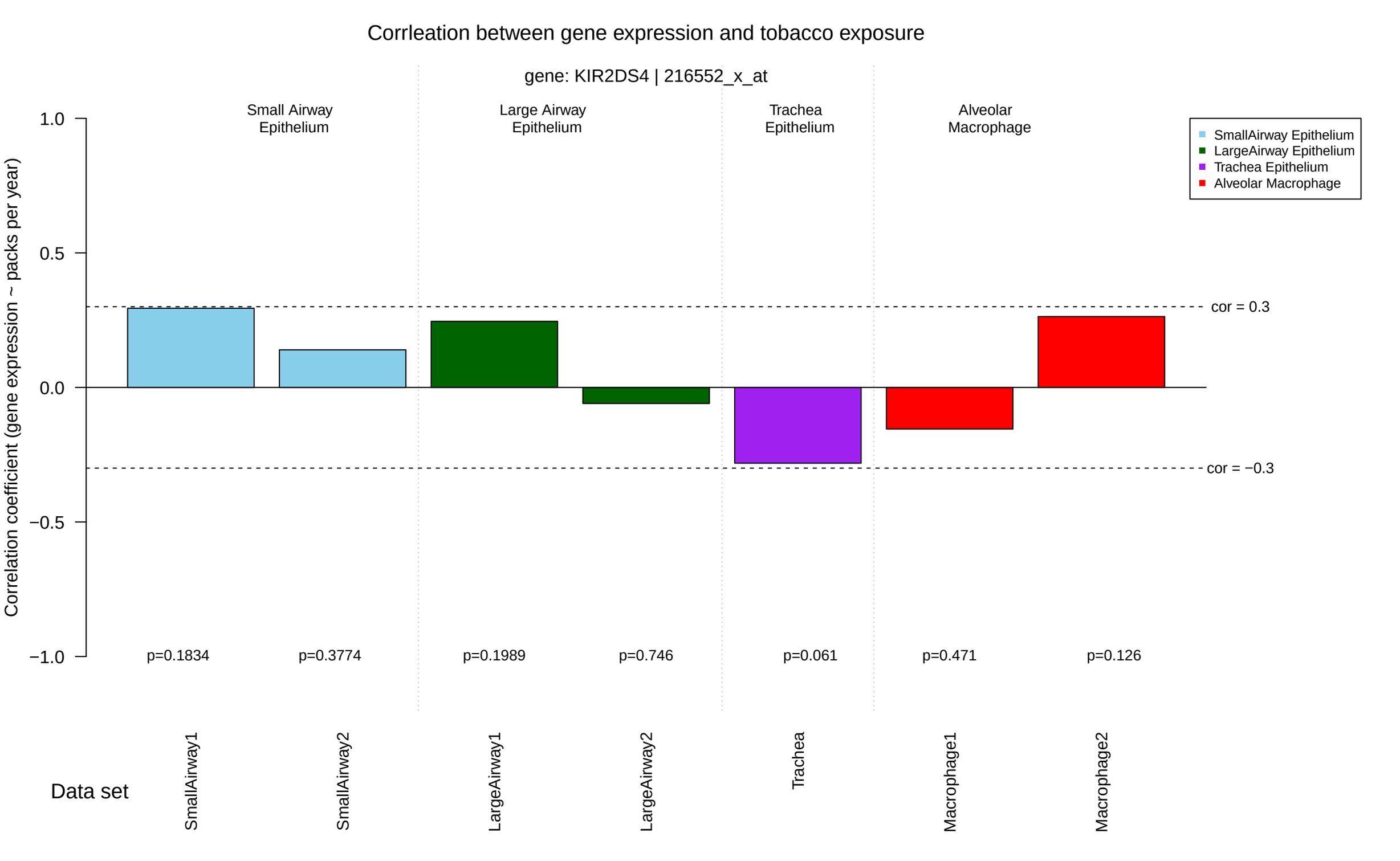

set: LargeAirway1 (p=0.7997)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6923)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2815)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0789)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2979)
KIR2DS4 | 216552_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KIR2DS4 | 216552_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1834
p=0.3774
p=0.1989
p=0.746
p=0.061
p=0.471
p=0.126
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2